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Project Report

On

**“Big Data Analytics of E-Commerce Sales Transactions Using
Hadoop MapReduce and Apache Pig”**

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Big Data Analytics of E-Commerce Sales Transactions

Using Hadoop MapReduce and Apache Pig

Abstract

E-commerce platforms generate continuous streams of transactional data covering orders, customers, products, payments, and delivery, and this volume only grows as a business scales. Unlike traditional retail, where a transaction happens face to face and much of the surrounding context is never recorded, an e-commerce order captures dozens of attributes automatically from device type and traffic source to delivery status and customer rating which makes the resulting datasets both larger and richer than their offline counterparts. This project applies the core Hadoop ecosystem the Hadoop Distributed File System (HDFS), MapReduce, and Apache Pig to a dataset of over one million e-commerce order records. HDFS is used to store the dataset in a distributed and fault-tolerant manner, splitting it into blocks across the cluster rather than relying on a single disk. Four MapReduce jobs, written in Java, compute foundational aggregates: total order count, category-wise sales, average order value by country, and total profit by payment method. Ten Apache Pig scripts then extend this analysis to cover payment distribution, monthly sales trends, best-selling products, top revenue countries, brand profitability, and delivery performance. Two of these analyses average order value by country were implemented independently in both MapReduce and Pig, allowing the results of one to be cross-checked against the other. The project demonstrates, end to end, how distributed storage and parallel processing can be used to turn a large raw transactional dataset into structured business insight, and compares the practical trade-offs between writing analysis logic by hand in Java MapReduce versus expressing the same logic in Pig Latin.

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1. Introduction

E-commerce platforms record every stage of a transaction — browsing, ordering, payment, and delivery — resulting in datasets that are too large and too varied for a single machine or a conventional relational database to process efficiently. As order volumes climb into the hundreds of thousands or millions, queries that would return instantly on a small spreadsheet start to take minutes or hours on a single machine, and the risk of losing data to a single point of failure becomes a genuine operational concern rather than a theoretical one. This project works with a dataset of order-level e-commerce records containing customer demographics, product details, pricing, payment method, shipping information, and delivery outcomes, giving a realistic picture of what a mid-sized online retailer's raw transaction log actually looks like in practice.

The objective is to apply distributed Big Data tools to this dataset in three complementary ways. First, HDFS is used to store the raw dataset reliably across the (pseudo-distributed) Hadoop cluster, so that the data survives node failure and can be read in parallel by multiple tasks at once. Second, MapReduce is used to implement four core aggregation jobs directly in Java, giving fine-grained control over exactly how each record is read, mapped, and reduced. Third, Apache Pig is used to perform ten further analyses using Pig Latin, which expresses the same style of group-by/aggregate logic far more concisely than hand-written MapReduce code, while still compiling down to MapReduce jobs behind the scenes.

This report documents the complete workflow environment setup, HDFS preparation, MapReduce implementation, and Pig analysis together with the exact commands used and the results obtained at each stage, so that the process can be reproduced or extended by anyone reading it. Screenshots referenced throughout are meant to be inserted at the marked points to provide visual proof that each step executed successfully.

2. Problem Statement

A single e-commerce order record is simple, but a merchant's real question is rarely about one order it is about patterns across a million of them: which categories sell best, which countries generate the most revenue, which payment methods are most profitable, and where delivery is under-performing. These are exactly the kinds of questions that decide inventory planning, marketing spend, and logistics partnerships in a real business, so answering them accurately matters well beyond the technical exercise itself. Answering these questions over a dataset of this size with a conventional single-machine script becomes slow and memory-constrained as the data grows, and the problem only worsens as more months of orders are added to the pile.

This project addresses that problem by distributing both storage and computation. The dataset is stored in HDFS so that it is split into blocks across the cluster rather than held on one disk, which also protects against data loss through replication across nodes. MapReduce and Apache Pig are then used to run the actual aggregation logic in parallel across those blocks, so that operations such as counting orders, summing revenue by category, or averaging order value by country scale with the cluster rather than with a single CPU. In this sense, the problem being solved is not just producing the right numbers once, but producing them in a way that keeps working as the dataset keeps growing.

3. Dataset Description

The dataset used is `ecommerce_dataset_+1m.csv`, containing just over one million e-commerce order records collected from Kaggle. Each row describes a single order and includes fields such as:

- Order details — `order_id`, `order_date`, `order_year/month/day/hour/minute/second`, `is_weekend`, `order_status`, `return_reason`
- Customer details — `customer_id`, `customer_name`, `gender`, `age`, `customer_segment`, `country`, `city`, `customer_loyalty_score`, `total_orders_by_customer`, `account_creation_date`
- Product details — `product_id`, `product_name`, `category`, `sub_category`, `brand`, `product_rating_avg`, `product_reviews_count`, `stock_quantity`
- Order economics — `unit_price_usd`, `quantity`, `discount_percent`, `discount_amount_usd`, `total_price_usd`, `cost_usd`, `profit_usd`, `tax_usd`, `currency`
- Payment and shipping — `payment_method`, `payment_status`, `installment_plan`, `shipping_method`, `shipping_cost_usd`, `delivery_days`, `shipping_country`, `warehouse_location`, `delivery_status`
- Customer experience — `rating`, `review_sentiment`, `customer_feedback`, `coupon_used`, `coupon_code`, `campaign_source`, `device_type`, `traffic_source`, `session_duration_minutes`, `pages_visited`, `abandoned_cart_before`, `fraud_risk_score`, `profit_margin_percent`, `order_priority`, `support_ticket_created`

Reviewing these fields beforehand was essential as they dictate the scope of MapReduce and Pig aggregations such as `category` and `total_price_usd` for category-sales, or `country` for average order values. Unused fields (e.g., `device_type`, `fraud_risk_score`) were retained in the schema to enable future analyses without needing to re-clean the dataset.

Dataset: <https://www.kaggle.com/datasets/akrambelha/global-e-commerce-dataset-1m-records-20242026>

4. Setting Up the Working Environment

Before any HDFS, MapReduce, or Pig work could begin, the required tools had to be installed and verified from the command line, since a missing or misconfigured tool at this stage would cause confusing failures much later in the pipeline:

- Java (JDK) — required since Hadoop and MapReduce programs run on the JVM.
- Hadoop 3.2.4 — verified with `hadoop version`, with `java -version` and `hadoop version` set so Hadoop commands run from any Command Prompt.
- Apache Pig 0.18.0 — verified with `pig -version`.

If all these commands return valid version information without errors, the environment is confirmed ready, and the project can proceed to starting the Hadoop services and preparing the dataset.

```
C:\Windows\system32>java -version
java version "1.8.0_411"
Java(TM) SE Runtime Environment (build 1.8.0_411-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.411-b09, mixed mode)

C:\Windows\system32>hadoop version
Hadoop 3.2.4
Source code repository Unknown -r 7e5d9983b388e372fe640f21f048f2f2ae6e9eba
Compiled by ubuntu on 2022-07-12T11:58Z
Compiled with protoc 2.5.0
From source with checksum ee031c16fe785bbb35252c749418712
This command was run using /C:/hadoop/share/hadoop/common/hadoop-common-3.2.4.jar

C:\Windows\system32>pig -version
Apache Pig version 0.18.0 (r1926484)
compiled Jun 16 2025, 10:38:21

C:\Windows\system32>
```

5. Storing the E-Commerce Data in HDFS

5.1 Launching the Hadoop Services

This step starts the Hadoop services so that HDFS and YARN are running and ready to store and process data.

HDFS and YARN were started so that the cluster could store and process data. Running `start-dfs.cmd` and `start-yarn.cmd` launches the required daemons, and `jps` confirms that NameNode, DataNode, ResourceManager, and NodeManager are all running before any further commands are attempted.

```
C:\hadoop\sbin>start-dfs.cmd

C:\hadoop\sbin>start-yarn.cmd
starting yarn daemons

C:\hadoop\sbin>jps
10132 NodeManager
10696 DataNode
280 Jps
9320 NameNode
4828 ResourceManager
```

If all these processes appear in the `jps` output, Hadoop is running correctly and the environment is ready to create HDFS directories and upload the dataset.

5.2 Create a Project Directory

A local project folder, `C:\ECommerceBigData`, was created with separate subfolders to keep the dataset, MapReduce source code, compiled classes, packaged JAR, and Pig scripts

organised and isolated from one another. Keeping these separated from the start avoided the kind of mix-up where a script or class file gets overwritten by accident later in the project.

```
C:\hadoop\sbin>cd C:\
C:\>mkdir ECommerceBigData
C:\>cd ECommerceBigData
C:\ECommerceBigData>mkdir dataset
C:\ECommerceBigData>mkdir java
C:\ECommerceBigData>mkdir classes
C:\ECommerceBigData>mkdir jar
C:\ECommerceBigData>mkdir pig
```

A corresponding HDFS directory structure, /ecommerce, was initialized to handle distributed data storage and separate input files from job outputs.

- **/ecommerce/input:** Stores the raw dataset uploaded for processing
- **/ecommerce/output:** Main container for all processing results
- **/ecommerce/output/mapreduce:** Dedicated storage for MapReduce job outputs
- **/ecommerce/output/pig:** Dedicated storage for Pig script outputs

```
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/input
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output/mapreduce
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output/pig
```

5. Dataset Upload and Check

To make the dataset available for distributed processing in Hadoop, the local file was uploaded to the Hadoop Distributed File System (HDFS)

File Presence Check: The command was executed on the C:\ECommerceBigData\dataset directory to verify that ecommerce_dataset_+1m.csv existed in the designated local folder.

```
C:\ECommerceBigData>dir C:\ECommerceBigData\dataset
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9
```

Integrity & Size Verification: The file size was checked (~410 MB) to confirm the dataset was complete and had not been corrupted or truncated before initiating the HDFS transfer.

```

C:\ECommerceBigData>dir C:\ECommerceBigData\dataset
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\dataset

09/10/2026  08:56 PM    <DIR>          .
09/10/2026  08:56 PM    <DIR>          ..
02/03/2026  11:28 AM             410,450,731  ecommerce_dataset_+1m.csv
               1 File(s)          410,450,731 bytes
               2 Dir(s)      80,965,586,944 bytes free

```

Confirming these details early prevents wasted effort later if the CSV were corrupted or malformed, every downstream HDFS, MapReduce, and Pig step would fail. Once the file is verified as complete and correctly structured, it's ready to be uploaded to HDFS.

5.4 Building the HDFS Directory Structure

The following commands set up the HDFS folder structure that separates raw input from the results generated later by MapReduce and Pig, mirroring the same separation already used in the local project folder:

```

C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/input
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output/mapreduce
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output/pig

```

Creating this structure in advance keeps the HDFS workspace organized and separates raw input data from generated output, which is especially important since multiple MapReduce jobs and Pig scripts.

5.5 Copying the Dataset into HDFS

This step transfers the dataset from the local file system into HDFS so that it can be accessed and processed in a distributed manner by MapReduce and Pig.

```

C:\ECommerceBigData>hdfs dfs -put "C:\ECommerceBigData\dataset\ecommerce_dataset_+1m.csv" /ecommerce/input/
C:\ECommerceBigData>hdfs dfs -ls /ecommerce/input
Found 1 items
-rw-r--r--  1 Anamikaa supergroup  410450731 2026-09-10 21:01 /ecommerce/input/ecommerce_dataset_+1m.csv

```

5.6 Verifying the Uploaded Dataset

This step confirms that the dataset uploaded to HDFS is complete and matches the original local file, ensuring no data was lost or corrupted during the transfer.

File Size Verification: The `hdfs dfs -du -h` command was run to check the human-readable file in HDFS, ensuring the dataset was fully transferred without data loss.

```
C:\ECommerceBigData\pig>hdfs dfs -du -h /ecommerce/input/ecommerce_dataset_+1m.csv
391.4 M  391.4 M  /ecommerce/input/ecommerce_dataset_+1m.csv
```

Record Count Validation: The command returned a line count of 1,000,124, confirming that the dataset contains 1,000,123 transactional records alongside the single CSV header row.

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/input/ecommerce_dataset_+1m.csv | find /c /v ""
1000124
```

Schema & Header Inspection: The `hdfs dfs -cat` command combined with PowerShell was used to read the first line of the file, confirming the dataset header and verifying all 62 columns were properly loaded into HDFS.

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/input/ecommerce_dataset_+1m.csv | powershell -Command "$input | Select-Object -First 1"
order_id,order_date,order_year,order_month,order_day,order_hour,order_minute,order_second,is_weekend,order_status,return_reason,customer_id,customer_name,gender,age,customer_segment,country,city,customer_loyalty_score,total_orders_by_customer,account_creation_date,product_id,product_name,category,sub_category,brand,product_rating_avg,product_reviews_count,stock_quantity,unit_price_usd,quantity,discount_percent,discount_amount_usd,total_price_usd,cost_usd,profit_usd,tax_usd,currency,payment_method,payment_status,installment_plan,shipping_method,shipping_cost_usd,delivery_days,shipping_country,warehouse_location,delivery_status,rating,review_sentiment,customer_feedback,coupon_used,coupon_code,campaign_source,device_type,traffic_source,session_duration_minutes,pages_visited,abandoned_cart_before,fraud_risk_score,profit_margin_percent,order_priority,support_ticket_created
cat: Unable to write to output stream.
```

6. Distributed Processing Using MapReduce

Four MapReduce jobs were implemented in Java, each with its own Driver, Mapper, and Reducer class, and packaged together into a single JAR (ecommerce.jar). Building all four into one JAR meant that only one file needed to be submitted to Hadoop for every job, rather than rebuilding a separate package each time a new analysis was added.

6.1 Writing the Driver, Mapper and Reducer Classes

For each job, Three Java classes were written: a Mapper, a Reducer, and a Driver. This separation follows the standard MapReduce pattern, where the Mapper handles per-record logic and the Reducer handles the cross-record aggregation, keeping each class focused on a single responsibility.

```
C:\ECommerceBigData>dir java
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\java

09/10/2026  09:07 PM    <DIR>          .
09/10/2026  09:07 PM    <DIR>          ..
09/10/2026  09:07 PM                965 Driver1.java
09/10/2026  09:06 PM                855 Mapper1.java
09/10/2026  09:06 PM                556 Reducer1.java
               3 File(s)            2,376 bytes
               2 Dir(s)  80,546,480,128 bytes free
```

6.2 Compile the MapReduce Classes Folder

The three Java files are compiled using `javac` (with the Hadoop classpath included). Compiling here catches any syntax or dependency errors before the job is packaged and run.

```

C:\ECommerceBigData>dir classes
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\classes

09/10/2026  09:10 PM    <DIR>          .
09/10/2026  09:10 PM    <DIR>          ..
09/10/2026  09:10 PM                1,349 Driver1.class
09/10/2026  09:10 PM                1,862 Mapper1.class
09/10/2026  09:10 PM                1,586 Reducer1.class
               3 File(s)                4,797 bytes
               2 Dir(s)  80,548,618,240 bytes free

```

6.3 Compiling the Java Source

Each class was compiled with javac, with the Hadoop common and MapReduce libraries included on the classpath so that Hadoop-specific types (such as Mapper, Reducer, and Job) would resolve correctly. Compiling at this stage, before packaging, made it possible to catch and fix syntax or type errors early rather than discovering them only when a job was submitted to the cluster.

```

C:\ECommerceBigData>hadoop classpath
C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\hadoop\hdfs\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\mapreduce\lib\*;C:\hadoop\share\hadoop\mapreduce\*

C:\ECommerceBigData>javac -classpath "%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" -d classes java\Mapper1.java java\Reducer1.java java\Driver1.java

```

6.4 Building the JAR File

All twelve compiled .class files (three per job, across four jobs) were packaged into a single ecommerce.jar using the jar command. This is the file that gets submitted to Hadoop with hadoop jar, since Hadoop runs MapReduce jobs from a packaged JAR rather than from loose .class files sitting in a folder.

```

C:\ECommerceBigData>cd classes

C:\ECommerceBigData\classes>jar -cvf ecommerce.jar *.class
added manifest
adding: Driver1.class(in = 1349) (out= 751)(deflated 44%)
adding: Mapper1.class(in = 1862) (out= 788)(deflated 57%)
adding: Reducer1.class(in = 1586) (out= 662)(deflated 58%)

```

```

C:\ECommerceBigData\classes>cd ..

C:\ECommerceBigData>dir classes\ecommerce.jar
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\classes

09/10/2026  09:23 PM                2,899 ecommerce.jar
               1 File(s)                2,899 bytes
               0 Dir(s)  80,320,385,024 bytes free

```

6.5 Run the MapReduce

Four MapReduce jobs were implemented they are:

- **Job 1-OrderCount** = Total Order Count
- **Job 2-CategorySales** = Category-wise Total Sales
- **Job 3-AvgOrderValue** = Average Order Value by Country
- **Job 4-ProfitByPaymentMethod** = Total Profit by Payment Method

Browse Directory

/ecommerce/output/mapreduce

Go!

Show

25

entries

Search:

☐		Permission		Owner		Group		Size		Last Modified		Replication		Block Size		Name	
☐		drwxr-xr-x		Anamikaa		supergroup		0 B		Sep 10 21:28		0		0 B		Job1-TotalOrderCount	
☐		drwxr-xr-x		Anamikaa		supergroup		0 B		Sep 10 21:48		0		0 B		Job2-CategoryWiseSales	
☐		drwxr-xr-x		Anamikaa		supergroup		0 B		Sep 10 21:51		0		0 B		Job3-AvgOrderValueByCountry	
☐		drwxr-xr-x		Anamikaa		supergroup		0 B		Sep 10 21:54		0		0 B		Job4-TotalProfitByPaymentMethod	

Showing 1 to 4 of 4 entries

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Next

6.5.1 Job 1-Counting Total Orders

This job counts the total number of valid order records in the dataset. The Mapper emits a constant key with a value of 1 for each order; the Reducer sums these to a single total. Although conceptually the simplest of the four jobs, it establishes a baseline figure that every later per-category or per-country percentage can be compared against.

6.5.1.1 Run the MapReduce job

The job is submitted using `hadoop jar`, pointing to the input dataset in HDFS and specifying an output directory for the results. The Mapper reads each record and emits a constant key with a count of 1 per valid order record, while the Reducer sums these counts to produce the final overall total.

```
C:\ECommerceBigData>hadoop jar classes\ecommerce.jar Driver1 /ecommerce/input/ecommerce_dataset_rim.csv /ecommerce/output/mapreduce/job1
2026-09-10 21:28:01,230 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:28:02,014 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:28:02,030 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamikaa/.staging/job_1789050973944_0001
2026-09-10 21:28:02,225 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 21:28:02,301 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:28:02,414 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0001
2026-09-10 21:28:02,416 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:28:02,579 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:28:02,580 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2026-09-10 21:28:02,991 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0001
2026-09-10 21:28:03,032 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0001/
2026-09-10 21:28:03,033 INFO mapreduce.Job: Running job: job_1789050973944_0001
2026-09-10 21:28:16,234 INFO mapreduce.Job: Job job_1789050973944_0001 running in uber mode : false
2026-09-10 21:28:16,237 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:28:24,408 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:28:30,481 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:28:35,601 INFO mapreduce.Job: Job job_1789050973944_0001 completed successfully
2026-09-10 21:28:35,605 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=19002343
  FILE: Number of bytes written=38959507
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=410459307
  HDFS: Number of bytes written=21
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=3
  Total time spent by all maps in occupied slots (ms)=17238
  Total time spent by all reduces in occupied slots (ms)=4273
  Total time spent by all map tasks (ms)=17238
```

6.5.1.2 Check the Result Directory

Once the job completes successfully, `hdfs dfs -ls /ecommerce/output/mapreduce/TotalOrderCount` is used to confirm the output directory was created properly, containing a `_SUCCESS` marker file

alongside the generated part-r-00000 result file.

```
C:\ECommerceBigData>hdfs dfs -ls /ecommerce/output/mapreduce/job1
Found 2 items
-rw-r--r--  1 Anamikaa supergroup      0 2026-09-10 21:28 /ecommerce/output/mapreduce/job1/_SUCCESS
-rw-r--r--  1 Anamikaa supergroup    21 2026-09-10 21:28 /ecommerce/output/mapreduce/job1/part-r-00000

C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job1/part-r-00000
Total_Orders    1000123
```

6.5.1.3 Display the Final Output

he final count is displayed using `hdfs dfs -cat /ecommerce/output/mapreduce/TotalOrderCount/part-r-00000`, showing the total number of order records in the dataset. This confirms that the MapReduce job correctly processed all records in the input CSV file and produced a single aggregated value across the dataset.

File contents

```
Total_Orders 1000123
```

6.5.2 Job 2-Sales Totals per Category

This job groups orders by product category and sums `total_price_usd` for each category, showing which product categories generate the most revenue. Because the Mapper emits category as the key and `total_price_usd` as the value, the Reducer's job reduces to a simple running sum per key, which Hadoop's shuffle-and-sort phase groups together automatically before the Reducer ever sees the data.

6.5.2.1 Run the MapReduce Job

The job is submitted using `hadoop jar`, pointing to the input dataset in HDFS and specifying an output directory for the results. The Mapper reads each record and emits the product category as the key and `total_price_usd` as the value, while the Reducer sums these amounts per category to compute total sales.

```

C:\ECommerceBigData>hadoop jar classes\ecommerce.jar Driver2 /ecommerce/input/ecommerce_dataset_1m.csv /ecommerce/output/mapreduce/job2
2026-09-10 21:47:50,799 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:47:51,549 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:47:51,560 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamika/.staging/job_1789050973944_0002
2026-09-10 21:47:51,752 INFO Input.FileInputFormat: Total input files to process : 1
2026-09-10 21:47:51,810 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:47:51,904 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0002
2026-09-10 21:47:51,906 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:47:52,088 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:47:52,088 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 21:47:52,141 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0002
2026-09-10 21:47:52,174 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0002/
2026-09-10 21:47:52,174 INFO mapreduce.Job: Running job: job_1789050973944_0002
2026-09-10 21:48:04,377 INFO mapreduce.Job: Job job_1789050973944_0002 running in uber mode : false
2026-09-10 21:48:04,380 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:48:13,586 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:48:19,046 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:48:24,726 INFO mapreduce.Job: Job job_1789050973944_0002 completed successfully
2026-09-10 21:48:24,804 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=18000102
    FILE: Number of bytes written=36955069
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=410459307
    HDFS: Number of bytes written=142
    HDFS: Number of read operations=14
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=3
    Launched reduce tasks=1
    Data-local map tasks=3
    Total time spent by all maps in occupied slots (ms)=20009
    Total time spent by all reduces in occupied slots (ms)=4344
    Total time spent by all map tasks (ms)=20009

```

6.5.2.2 Check the Result Directory

Once the job completes successfully and this command is used to confirm the output directory was created properly, containing a `_SUCCESS` marker file alongside the generated `part-r-00000` result file.

```

C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job2/part-r-00000
Clothing      4.52057958999999134E7
Electronics   1.3986598693999827E8
Health        5.746792554999967E7
Home          8.782867776000017E7
Sports        7.291747085000059E7

```

6.5.2.3 Display the Final Output

The final aggregated result is displayed using command, showing the total sales generated by each product category. This confirms that the MapReduce job correctly processed all records in the input CSV file and mapped revenue figures across product categories.

File contents

```

Clothing 4.52057958999999134E7
Electronics 1.3986598693999827E8
Health 5.746792554999967E7
Home 8.782867776000017E7
Sports 7.291747085000059E7

```

6.5.3 Job 3-Average Order Value per Country

This job groups orders by country and computes the average total `_price_usd` per country, revealing whether spending patterns differ across markets. Computing a correct distributed

average required the Reducer to track both a running sum and a running count for each country, rather than averaging partial averages, since the latter would produce a mathematically incorrect result whenever countries have unequal numbers of orders.

6.5.3.1 Run the MapReduce Job

The job is submitted using command pointing to the input dataset in HDFS and specifying an output directory for the results. The Mapper emits country as the key and total_price_usd as the value, while the Reducer tracks both running sums and running counts for each country to calculate an accurate average order value.

```
C:\ECommerceBigData>hadoop jar classes\ecommerce.jar Driver3 /ecommerce/input/ecommerce_dataset_1m.csv /ecommerce/output/mapreduce/job3
2026-09-10 21:51:05,128 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:51:05,910 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:51:05,924 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamikaa/.staging/job_1789050973944_0003
2026-09-10 21:51:06,109 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 21:51:06,168 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:51:06,266 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0003
2026-09-10 21:51:06,268 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:51:06,429 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:51:06,429 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 21:51:06,482 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0003
2026-09-10 21:51:06,515 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0003/
2026-09-10 21:51:06,516 INFO mapreduce.Job: Running job: job_1789050973944_0003
2026-09-10 21:51:18,736 INFO mapreduce.Job: Job job_1789050973944_0003 running in uber mode : false
2026-09-10 21:51:18,737 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:51:26,881 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:51:33,963 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:51:39,045 INFO mapreduce.Job: Job job_1789050973944_0003 completed successfully
2026-09-10 21:51:39,127 INFO mapreduce.Job: Counters: 54
    File System Counters
      FILE: Number of bytes read=19303086
      FILE: Number of bytes written=39561057
      FILE: Number of read operations=0
      FILE: Number of large read operations=0
      FILE: Number of write operations=0
      HDFS: Number of bytes read=410459307
      HDFS: Number of bytes written=278
      HDFS: Number of read operations=14
      HDFS: Number of large read operations=0
      HDFS: Number of write operations=2
      HDFS: Number of bytes read erasure-coded=0
    Job Counters
      Launched map tasks=3
      Launched reduce tasks=1
      Data-local map tasks=3
      Total time spent by all maps in occupied slots (ms)=18949
      Total time spent by all reduces in occupied slots (ms)=4304
      Total time spent by all map tasks (ms)=18949
```

6.5.3.2 Check the Result Directory

Once the job completes successfully, command is used to confirm the output directory was created properly, containing a _SUCCESS marker file alongside the generated part-r-00000 result file.

```
C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job3/part-r-00000
Australia      404.2269420891208
Belgium 402.7290564684118
Canada 401.9354866380752
France 401.71363695367967
Germany 402.63543595609747
Italy 405.7300347831337
Netherlands 403.24070583203815
Spain 403.6074160594136
United Kingdom 403.14696014550543
United States 403.40777806741994
```

6.5.3.3 Display the Final Output

The final aggregated result is displayed using command showing the average order value (AOV) in USD across all 10 countries. This confirms that the MapReduce job correctly

calculated distributed averages without precision loss across varying order counts per market.

File contents	
Australia	404.2269420891208
Belgium	402.7290564684118
Canada	401.9354866380752
France	401.71363695367967
Germany	402.63543595609747
Italy	405.7300347831337
Netherlands	403.24070583203815
Spain	403.6074160594136
Spain	403.6074160594136
United Kingdom	403.14696014550543
United States	403.40777806741994

6.5.4 Job 4-Profit Totals per Payment Method

This job groups orders by payment_method and sums profit_usd, showing whether any single payment method dominates overall profitability. This is a useful complement to Job 2's category-based view, since it looks at the same set of orders through the lens of how customers paid rather than what they bought, surfacing a completely different kind of business pattern from the same underlying data.

6.5.4.1 Run the MapReduce Job

The job is submitted using, command pointing to the input dataset in HDFS and specifying an output directory for the results. The Mapper reads each record and emits payment_method as the key and profit_usd as the value, while the Reducer sums the total profit for each payment option.

```
C:\ECommerceBigData\hadoop jar classes\ecommerce.jar Driver4 /ecommerce/input/ecommerce_dataset_1m.csv /ecommerce/output/mapreduce/job4
2026-09-10 21:54:14,188 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:54:14,934 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:54:14,948 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamika/.staging/job_1789050973944_0004
2026-09-10 21:54:15,129 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 21:54:15,187 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:54:15,283 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0004
2026-09-10 21:54:15,285 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:54:15,445 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:54:15,446 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 21:54:15,500 INFO impl.VarnClientImpl: Submitted application application_1789050973944_0004
2026-09-10 21:54:15,533 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8080/proxy/application_1789050973944_0004/
2026-09-10 21:54:15,534 INFO mapreduce.Job: Running job: job_1789050973944_0004
2026-09-10 21:54:27,737 INFO mapreduce.Job: Job job_1789050973944_0004 running in uber mode : false
2026-09-10 21:54:27,740 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:54:35,886 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:54:42,963 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:54:48,055 INFO mapreduce.Job: Job job_1789050973944_0004 completed successfully
2026-09-10 21:54:48,138 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=20883108
  FILE: Number of bytes written=42561485
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=418459387
  HDFS: Number of bytes written=159
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=3
```

6.5.4.2 Check the Result Directory

Once the job completes successfully, command is used to confirm the output directory was created properly, containing a `_SUCCESS` marker file alongside the generated `part-r-00000` result file.

```
C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job4/part-r-00000
Apple Pay      3.2072775629999757E7
Bank Transfer  3.2250553240000267E7
Credit Card   3.2119889630000137E7
Debit Card     3.2356712459999777E7
PayPal         3.2146881920000304E7
```

6.5.4.3 Display the Final Output

The final aggregated result is displayed using command showing the overall profit generated per payment method. This confirms that the MapReduce job correctly aggregated the profit figures to analyze profitability across payment types.

File contents

```
Apple Pay      3.2072775629999757E7
Bank Transfer  3.2250553240000267E7
Credit Card   3.2119889630000137E7
Debit Card     3.2356712459999777E7
PayPal         3.2146881920000304E7
```

6.5.5 Managing Java Source Files and Directory Structure

Directory listing of the java folder showing Java source files for all four MapReduce jobs.

```
C:\ECommerceBigData>dir java
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\java

09/10/2026  09:43 PM    <DIR>          .
09/10/2026  09:43 PM    <DIR>          ..
09/10/2026  09:07 PM             965 Driver1.java
09/10/2026  09:38 PM             921 Driver2.java
09/10/2026  09:42 PM             926 Driver3.java
09/10/2026  09:43 PM             926 Driver4.java
09/10/2026  09:06 PM             855 Mapper1.java
09/10/2026  09:36 PM             834 Mapper2.java
09/10/2026  09:40 PM             817 Mapper3.java
09/10/2026  09:42 PM             821 Mapper4.java
09/10/2026  09:06 PM             556 Reducer1.java
09/10/2026  09:37 PM             494 Reducer2.java
09/10/2026  09:41 PM             595 Reducer3.java
09/10/2026  09:43 PM             494 Reducer4.java
                12 File(s)              9,204 bytes
                2 Dir(s)  80,406,986,752 bytes free
```

Compiling the Java Source

```
C:\ECommerceBigData>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" -d classes java\Mapper2.java java\Reducer2.java java\Driver2.java java\Mapper3.java java\Reducer3.java java\Driver3.java java\Mapper4.java java\Reducer4.java java\Driver4.java
```

Building the JAR File

```
C:\ECommerceBigData>cd classes
C:\ECommerceBigData\classes>jar -cvf ecommerce.jar *.class
added manifest
adding: Driver1.class(in = 1349) (out= 751)(deflated 44%)
adding: Driver2.class(in = 1360) (out= 760)(deflated 44%)
adding: Driver3.class(in = 1365) (out= 763)(deflated 44%)
adding: Driver4.class(in = 1365) (out= 764)(deflated 44%)
adding: Mapper1.class(in = 1862) (out= 788)(deflated 57%)
adding: Mapper2.class(in = 1903) (out= 796)(deflated 58%)
adding: Mapper3.class(in = 1903) (out= 796)(deflated 58%)
adding: Mapper4.class(in = 1903) (out= 796)(deflated 58%)
adding: Reducer1.class(in = 1586) (out= 662)(deflated 58%)
adding: Reducer2.class(in = 1604) (out= 663)(deflated 58%)
adding: Reducer3.class(in = 1643) (out= 697)(deflated 57%)
adding: Reducer4.class(in = 1604) (out= 664)(deflated 58%)
```

7. Extended Analysis Using Apache Pig

Apache Pig was used to extend the analysis beyond the four MapReduce jobs, using Pig Latin scripts that load, clean, group, and aggregate the same dataset with considerably less code than the equivalent Java would require. All scripts were stored under the local pig folder and run with `pig -x mapreduce <script>.pig`, with results stored in HDFS under `/ecommerce/output/pig`. Because Pig compiles each script down into one or more MapReduce jobs automatically, the underlying execution model is identical only the amount of code the analyst has to write by hand changes.

7.1 Create the Pig Script Folder

This step sets up the dedicated Pig workspace and organizes it for the full set of analyses to come.

```
C:\ECommerceBigData>cd C:\ECommerceBigData\pig
C:\ECommerceBigData\pig>notepad pig\pig1_rowcheck.pig
C:\ECommerceBigData\pig>notepad pig1_rowcheck.pig
```

All ten scripts sit directly inside the pig folder, distinguished by descriptive, numbered file names:

- `pig1_rowcheck.pig` — data loading and cleaning (row count check)
- `pig2_ordercount.pig` — total order count

- pig3_categorysales.pig — category-wise sales
- pig4_countryavg.pig — average order value by country
- pig5_paymentdist.pig — payment method distribution
- pig6_monthlytrend.pig — monthly sales trend
- pig7_topproducts.pig — top 10 best-selling products
- pig8_topcountries.pig — top 10 countries by revenue
- pig9_brandprofit.pig — brand-wise total profit
- pig10_deliverystatus.pig — delivery status breakdown

Browse Directory

/ecommerce/output/pig

Go!



Show

25

entries

Search:

☐	 Permission	 Owner	 Group	 Size	 Last Modified	 Replication	 Block Size	 Name	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:13	0	0 B	AvgOrderValueByCountry	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 15:13	0	0 B	BrandWiseTotalProfit	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 13:59	0	0 B	CategoryWiseSales	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 15:17	0	0 B	DeliveryStatusBreakdown	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:19	0	0 B	PaymentMethodDistribution	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 13:43	0	0 B	RowCountCheck	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:25	0	0 B	RowCountCheckMonthlySalesTrend	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:57	0	0 B	Top10BestSellingProducts	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 15:07	0	0 B	Top10CountriesByRevenue	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 13:52	0	0 B	TotalOrderCount	

Showing 1 to 10 of 10 entries

Previous

1

Next

Showing 1 to 10 of 10 entries

Previous 1 Next

7.2 Pig analysis

7.2.1 Pig Data Loading & Cleaning

Every script begins by loading the CSV with an explicit schema and filtering out the header row and records missing an `order_id` or `total_price_usd`. A dedicated check script (pig1_rowcheck.pig) confirms the raw versus cleaned row counts, acting as a sanity check that is run once before trusting the other nine analyses.

7.2.1.1 Create the Pig file and Run the Pig script

```

C:\ECommerceBigData\pig>pig -x mapreduce pig1_rowcheck.pig
2026-09-11 13:12:49,628 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 13:12:49,631 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 13:12:49,631 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 13:12:50,317 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 13:12:50,318 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789110770306.log
2026-09-11 13:12:51,299 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER7\pigbootstrap not found
2026-09-11 13:12:51,397 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 13:12:51,397 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 13:12:52,508 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig1_rowcheck.pig-b19eecd1-4c2d-4506-89c2-f9996f07ff0e
2026-09-11 13:12:52,508 [main] WARN org.apache.pig.PigServer - ATs is disabled since yarn.timeline-service.enabled set to false
2026-09-11 13:12:54,091 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-09-11 13:12:54,129 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 13:12:54,194 [main] INFO org.apache.pig.newplan.Logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 13:12:54,325 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 13:12:54,467 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 13:12:54,501 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 13:12:54,558 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 13:12:54,558 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 13:12:54,738 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:12:55,270 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 13:12:55,290 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 13:12:55,305 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 13:12:55,306 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 13:12:55,312 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 13:12:55,319 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 13:12:55,321 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 13:12:55,325 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 13:12:55,326 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 13:12:55,361 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 13:12:55,647 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp572521400/tmp2022932997/pig-0.18.0-core-h3.jar

```

7.2.1.2 Show the actual output

```

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/analysis1/part-m-00000
Cleaned Row Count,1000123

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/analysis1/part-m-00001
Raw Row Count,1000124

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/analysis1/part-*
Cleaned Row Count,1000123
Raw Row Count,1000124

```

File contents	
Cleaned Row Count,1000123	
Raw Row Count,1000124	

7.2.2 Counting Total Orders in Pig

Groups all cleaned records together and counts them, giving the same figure as MapReduce Job 1 but expressed in three lines of Pig Latin instead of a full Java job with three separate classes. Comparing this figure against Job 1's output is a quick way to confirm both pipelines are reading and filtering the same data identically.

7.2.2.1 Create the Pig script

```

C:\ECommerceBigData\pig>notepad pig2_ordercount.pig

```

7.2.2.2 Run the Pig script

```

C:\ECommerceBigData\pig>pig -x mapreduce pig2_ordercount.pig
2026-09-11 13:51:31,461 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 13:51:31,463 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 13:51:31,463 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 13:51:32,135 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 13:51:32,135 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789113092127.log
2026-09-11 13:51:33,137 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 13:51:33,208 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 13:51:33,918 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig2_ordercount.pig-8edd8917-06fc-426d-9673-06d2ad636ccb
2026-09-11 13:51:33,919 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 13:51:35,011 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
textoutputformat.separator
2026-09-11 13:51:35,032 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY_FILTER
2026-09-11 13:51:35,059 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... Will not generate code.
2026-09-11 13:51:35,098 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalcul
ator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 13:51:35,173 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold
 = 489580128, usageThreshold = 489580128
2026-09-11 13:51:35,264 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 13:51:35,296 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 13:51:35,342 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 13:51:35,342 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 13:51:35,477 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:51:35,765 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, us
e yarn.system-metrics-publisher.enabled
2026-09-11 13:51:35,785 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 13:51:35,793 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-09-11 13:51:35,793 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
set, set to default 0.3
2026-09-11 13:51:35,796 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputfo
rmat.compress
2026-09-11 13:51:35,800 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required r
educers.
2026-09-11 13:51:35,801 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 13:51:35,801 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 13:51:35,802 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 13:51:35,824 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.fi
le.replication

```

7.2.2.3 Show the actual output

```

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/analysis2/part-r-00000
1000123

```

File contents

```
1000123
```

7.2.3 Sales Totals per Category in Pig

Groups records by category and computes both the order count and total sales per category using COUNT and SUM in a single FOREACH statement. Including both metrics side by side, rather than just total sales, makes it possible to see whether a category's high revenue comes from a large number of small orders or a smaller number of high-value ones.

7.2.3.1 Run the Pig script

```

C:\ECommerceBigData\pig>notepad pig3_categorysales.pig

C:\ECommerceBigData\pig>pig -x mapreduce pig3_categorysales.pig
2026-09-11 13:57:55,521 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 13:57:55,523 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 13:57:55,523 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 13:57:56,344 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 13:57:56,344 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789113476328.log
2026-09-11 13:57:57,440 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER/.pigbootup not found
2026-09-11 13:57:57,650 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 13:57:58,588 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 13:57:58,588 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig3_categorysales.pig-30e9ff89-d775-4241-bb60-33ebc20378da
2026-09-11 13:58:00,123 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline.service.enabled set to false
2026-09-11 13:58:00,123 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
extoutputformat.separator
2026-09-11 13:58:00,157 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 13:58:00,200 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 13:58:00,262 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalcul
ator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 13:58:00,376 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold
= 489580128, usageThreshold = 489580128
2026-09-11 13:58:00,457 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 13:58:00,507 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 13:58:00,555 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 13:58:00,555 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 13:58:00,695 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:58:00,695 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, us
e yarn.system-metrics-publisher.enabled
2026-09-11 13:58:01,079 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 13:58:01,088 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-09-11 13:58:01,088 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
set, set to default 0.3
2026-09-11 13:58:01,091 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutpuf
mat.compress
2026-09-11 13:58:01,095 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required r
educers.
2026-09-11 13:58:01,097 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.h
adoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 13:58:01,107 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=100000000 maxReducers=999
totalInputFileSize=410450731

2026-09-11 13:59:29,578 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:59:29,582 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to j
ob history server
2026-09-11 13:59:29,625 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:59:29,635 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to j
ob history server
2026-09-11 13:59:29,680 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:59:29,688 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to j
ob history server
2026-09-11 13:59:29,735 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSIO
N_FAILED 27 time(s).
2026-09-11 13:59:29,736 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 13:59:29,804 [main] INFO org.apache.pig.Main - Pig script completed in 1 minute, 34 seconds and 333 milliseconds (94333 ms)

```

7.2.3.2 Show the actual output

```

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/CategoryWiseSales/part-r-00000
Home,200589,8.782867776000027E7
Health,199574,5.746792554999983E7
Sports,200397,7.291747085000002E7
Clothing,199544,4.5205795900000416E7
Electronics,200019,1.398659869399997E8

```

File contents

```

Home,200589,8.782867776000027E7
Health,199574,5.746792554999983E7
Sports,200397,7.291747085000002E7
Clothing,199544,4.5205795900000416E7
Electronics,200019,1.398659869399997E8

```

7.2.4 Average Order Value by Country in Pig

Groups records by country and applies AVG on total_price_usd — the Pig equivalent of MapReduce Job 3, cross-checked against the MapReduce result for consistency. Because Pig's built-in AVG function handles the sum-and-count logic internally, this analysis needed

only a GROUP BY and a single FOREACH, considerably shorter than the equivalent Java Reducer.

7.2.4.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData\pig>notepad pig4_countryavg.pig

C:\ECommerceBigData\pig>pig -x mapreduce pig4_countryavg.pig
2026-09-11 14:12:31,067 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:12:31,070 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:12:31,071 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:12:31,966 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:12:31,966 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1789114351953.log
2026-09-11 14:12:33,182 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 14:12:33,251 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:12:33,251 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:12:33,957 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig4_countryavg.pig-e6e23500-a853-41ed-9113-bc06e24e6783
2026-09-11 14:12:33,957 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:12:35,150 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputtextoutputformat.separator
2026-09-11 14:12:35,171 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 14:12:35,203 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 14:12:35,259 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:12:35,378 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 14:12:35,500 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:12:35,528 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:12:35,559 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 14:12:35,560 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 14:12:35,600 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:12:35,953 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 14:12:35,970 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:12:35,978 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 14:12:35,979 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 14:12:35,986 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 14:12:35,990 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 14:12:35,992 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 14:12:36,003 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=410450731
```

7.2.4.2 Show the actual output

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/AvgOrderValueByCountry/part-r-00000
Italy,405.7300347831321
Spain,403.60741605941524
Canada,401.9354866380731
France,401.7136369536785
Belgium,402.72905646840763
Germany,402.63543595609656
Australia,404.22694208912384
Netherlands,403.2407058320406
United States,403.4077780674112
United Kingdom,403.14696014550645
```

File contents

```
Italy,405.7300347831321
Spain,403.60741605941524
Canada,401.9354866380731
France,401.7136369536785
Belgium,402.72905646840763
Germany,402.63543595609656
Australia,404.22694208912384
Netherlands,403.2407058320406
```

7.2.5 Payment Method Distribution in Pig

Groups records by payment_method and computes both order count and total revenue per method, giving a fuller picture than the profit-only view from MapReduce Job 4. Looking

at revenue alongside profit for the same payment methods helps distinguish payment methods that are simply popular from ones that are also disproportionately profitable.

7.2.5.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData\pig>notepad pig5_paymentdist.pig

C:\ECommerceBigData\pig>pig -x mapreduce pig5_paymentdist.pig
2026-09-11 14:18:42,611 [main] INFO org.apache.pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:18:42,614 [main] INFO org.apache.pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:18:42,615 [main] INFO org.apache.pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:18:43,469 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:18:43,470 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789114723455.log
2026-09-11 14:18:44,671 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER\pigbootstrap not found
2026-09-11 14:18:44,752 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:18:44,752 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:18:45,479 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig5_paymentdist.pig-e5109fe5-65ce-4b02-b3ae-78ffe8a5c074
2026-09-11 14:18:45,479 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:18:46,612 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputtextoutputformat.separator
2026-09-11 14:18:46,634 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 14:18:46,662 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 14:18:46,701 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED:[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:18:46,776 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 14:18:46,879 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:18:46,920 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:18:46,980 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 14:18:46,981 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 14:18:47,138 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:18:47,398 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 14:18:47,414 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:18:47,421 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 14:18:47,423 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 14:18:47,425 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 14:18:47,429 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
```

7.2.5.2 Show the actual output

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/PaymentMethodDistribution/part-r-00000
PayPal,199881,8.05951279600001E7
Apple Pay,199814,8.039556929000005E7
Debit Card,200805,8.106516731000024E7
Credit Card,199410,8.041428974000017E7
Bank Transfer,200213,8.081570269999954E7
```

File contents

```
PayPal,199881,8.05951279600001E7
Apple Pay,199814,8.039556929000005E7
Debit Card,200805,8.106516731000024E7
Credit Card,199410,8.041428974000017E7
Bank Transfer,200213,8.081570269999954E7
```

7.2.6 Monthly Sales Trend in Pig

Groups records by order_month, computes order count and total sales per month, and orders the result chronologically using ORDER BY to reveal seasonal demand patterns. Sorting the output this way was necessary because Pig does not guarantee any particular ordering of grouped results by default, and a chronological view is essential for spotting a genuine trend rather than an arbitrary one.

7.2.6.1 Create the Pig file and Run the Pig script

```

C:\ECommerceBigData\pig>notepad pig6_monthlytrend.pig
C:\ECommerceBigData\pig>pig -x mapreduce pig6_monthlytrend.pig
2026-09-11 14:22:11,520 [main] INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:22:11,521 [main] INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:22:11,523 [main] INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:22:12,163 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:22:12,164 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789114932148.log
2026-09-11 14:22:13,245 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 14:22:13,314 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:22:13,314 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:22:14,138 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig6_monthlytrend.pig-0b22c2d5-e7de-4a73-b080-0534ead47ac5
2026-09-11 14:22:14,138 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:22:15,321 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputformat.separator
2026-09-11 14:22:15,343 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-11 14:22:15,373 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 14:22:15,429 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergerForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlattem, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:22:15,544 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699480192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 14:22:15,679 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:22:15,764 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:22:15,796 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-220
2026-09-11 14:22:15,816 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-11 14:22:15,816 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-11 14:22:15,956 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:22:16,251 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 14:22:16,267 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:22:16,276 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 14:22:16,277 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 14:22:16,279 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress

```

7.2.6.2 Show the actual output

```

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/MonthlySalesTrend2/part-r-00000
1,85100,3.44101542699999936E7
2,77685,3.1398028690000005E7
3,85079,3.418629474000016E7
4,82436,3.33436393699999915E7
5,85100,3.4282929650000095E7
6,81699,3.2865246930000104E7
7,85029,3.433948877999994E7
8,84799,3.4242546549999945E7
9,82005,3.3027247350000087E7
10,84848,3.4253045459999874E7
11,81731,3.286898035000004E7
12,84612,3.406825486000006E7

```

File contents

```

1,85100,3.44101542699999936E7
2,77685,3.1398028690000005E7
3,85079,3.418629474000016E7
4,82436,3.33436393699999915E7
5,85100,3.4282929650000095E7
6,81699,3.2865246930000104E7
7,85029,3.433948877999994E7
8,84799,3.4242546549999945E7

```

7.2.7 Identifying the Top-Selling Products

Groups records by product_name, sums quantity sold, sorts in descending order, and limits to the top 10 — identifying the products driving the most unit sales. This view is deliberately based on quantity rather than revenue, so that a high-volume, low-price product is not overshadowed by a low-volume, high-price one in the same category.

7.2.7.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData\pig>notepad pig7_topproducts.pig
C:\ECommerceBigData\pig>pig -x mapreduce pig7_topproducts.pig
2026-09-11 14:52:35,784 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:52:35,786 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:52:35,787 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:52:36,700 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:52:36,700 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789116756689.log
2026-09-11 14:52:37,736 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 14:52:37,819 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:52:37,820 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:52:38,708 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig7_topproducts.pig-7552e280-4ac2-4e96-8752-acf53d28b464
2026-09-11 14:52:38,709 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:52:40,202 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputformat.separator
2026-09-11 14:52:40,235 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-11 14:52:40,270 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 14:52:40,315 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:52:40,403 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 14:52:40,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.mapreduce.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:52:40,658 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:52:40,694 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.mapreduce.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope=217
2026-09-11 14:52:40,714 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.mapreduce.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-11 14:52:40,715 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.mapreduce.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-11 14:52:40,867 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:52:41,285 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 14:52:41,306 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:52:41,317 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 14:52:41,317 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce.mapreduce.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 14:52:41,322 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
```

7.2.7.2 Show the actual output

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/Top10BestSellingProductsPig/part-r-00000
Water Bottle Insulated,67853
Vacuum Robot Cleaner,67703
Tennis Balls Pack,67482
Cotton T-Shirt Pack,67219
Gym Gloves,67215
Storage Box Set,67130
LED Strip Lights 5m,67124
Treadmill Folding,67109
Resistance Bands Set,67042
Desk Lamp LED,66928
```

File contents

```
Water Bottle Insulated,67853
Vacuum Robot Cleaner,67703
Tennis Balls Pack,67482
Cotton T-Shirt Pack,67219
Gym Gloves,67215
Storage Box Set,67130
LED Strip Lights 5m,67124
Treadmill Folding,67109
```

7.2.8 Identifying the Highest-Revenue Countries

Groups records by country, sums total_price_usd, sorts in descending order, and limits to the top 10 identifying the highest-revenue markets. This complements the per-country average, since a country can rank highly here simply by having a large number of orders even if its average order value is unremarkable.

7.2.8.1 Create the Pig file and Run the Pig script

```

C:\ECommerceBigData\pig>notepad pig8_topcountries.pig

C:\ECommerceBigData\pig>pig -x mapreduce pig8_topcountries.pig
2026-09-11 15:02:12,016 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 15:02:12,018 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 15:02:12,018 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 15:02:12,769 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 15:02:12,769 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789117332754.log
2026-09-11 15:02:13,989 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 15:02:14,062 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 15:02:14,062 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 15:02:14,783 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG:pig8_topcountries.pig-83e6b78f-a5bd-45ec-8d17-39b1b69235d0
2026-09-11 15:02:14,784 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 15:02:15,880 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputformat.separator
2026-09-11 15:02:15,900 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-11 15:02:15,932 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 15:02:15,991 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - (RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCasterInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCasterInserter])
2026-09-11 15:02:16,103 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 15:02:16,178 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 15:02:16,231 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 15:02:16,259 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce code scope-217
2026-09-11 15:02:16,276 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-11 15:02:16,277 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-11 15:02:16,419 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 15:02:16,709 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 15:02:16,730 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 15:02:16,738 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 15:02:16,738 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 15:02:16,742 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 15:02:16,745 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.

```

7.2.8.2 Show the actual output

```

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/Top10CountriesByRevenue/part-r-00000
Netherlands,4.0545046490000002E7
Italy,4.04760340000000045E7
Belgium,4.041667991999999E7
Spain,4.037728950999999E7
United Kingdom,4.034049741999999E7
Australia,4.033133892000024E7
United States,4.0236698599999726E7
France,4.023041560000004E7
Canada,4.0202793179999985E7
Germany,4.012906336000032E7

```

File contents

```

Netherlands,4.0545046490000002E7
Italy,4.04760340000000045E7
Belgium,4.041667991999999E7
Spain,4.037728950999999E7
United Kingdom,4.034049741999999E7
Australia,4.033133892000024E7
United States,4.0236698599999726E7
France,4.023041560000004E7

```

7.2.9 Brand Profitability Ranking

Groups records by brand and sums profit_usd, sorted in descending order, to identify which brands are most profitable overall as opposed to highest-selling. A brand that sells fewer units at a much higher margin can outrank a higher-volume brand here, which is a distinction the category-level view in Section 7.3 cannot show on its own.

7.2.9.1 Create the Pig file and Run the Pig script

```

C:\ECommerceBigData\pig>notepad pig9_brandprofit.pig
C:\ECommerceBigData\pig>pig -x mapreduce pig9_brandprofit.pig
2026-09-11 15:10:08,068 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 15:10:08,070 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 15:10:08,071 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 15:10:08,965 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 15:10:08,966 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789117888956.log
2026-09-11 15:10:10,199 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 15:10:10,175 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 15:10:10,175 [main] INFO org.apache.pig.backend.hadoop.executionengine.MExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 15:10:10,968 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig9_brandprofit.pig-9b7a7a3e-f38d-4eab-b25b-8cbfec025f8
2026-09-11 15:10:10,969 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 15:10:12,272 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
extoutputformat.separator
2026-09-11 15:10:12,295 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-11 15:10:12,325 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 15:10:12,374 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - (RULES ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalcul
ator, ForEachConstantCalculator, GroupByConstParallelSatter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter])
2026-09-11 15:10:12,463 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold
= 489580128, usageThreshold = 489580128
2026-09-11 15:10:12,543 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 15:10:12,596 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 15:10:12,625 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapRedu
ce node scope: 217
2026-09-11 15:10:12,644 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-11 15:10:12,645 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-11 15:10:12,778 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 15:10:13,047 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, us
e yarn.system-metrics-publisher.enabled
2026-09-11 15:10:13,065 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 15:10:13,073 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-09-11 15:10:13,074 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
set, set to default 0.3
2026-09-11 15:10:13,077 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputfo
rmat.compress
2026-09-11 15:10:13,080 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required r
educers.
2026-09-11 15:10:13,082 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.h
adoop.executionengine.mapReduceLayer.InputSizeReducerEstimator

```

7.2.9.2 Show the actual output

```

C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/BrandWiseTotalProfit/part-r-00000
Samsung,8136060.920000006
Dell,8109250.999999991
Fitbit,8102206.139999992
Sony,8096025.210000005
GoPro,8083747.099999981
JBL,8074264.540000049
Philips,8067932.990000002
Xiaomi,8066665.4
Baseus,8064639.049999995
Huawei,8057126.309999998
Anker,8046326.790000007
Nike,8038814.440000014
Apple,8012777.900000026
Logitech,8012292.110000023
Nikon,8004462.250000009
Adidas,8002090.960000033
Canon,7998829.029999991
HP,7997523.510000008
Lenovo,7994543.669999995
Asus,7981233.5600000145

```

File contents

```

Samsung,8136060.920000006
Dell,8109250.999999991
Fitbit,8102206.139999992
Sony,8096025.210000005
GoPro,8083747.099999981
JBL,8074264.540000049
Philips,8067932.990000002
Xiaomi,8066665.4

```

7.2.10 Delivery Performance Breakdown

Groups records by delivery_status and computes order count alongside the average customer rating for each status, connecting delivery performance to customer satisfaction.

This is the one analysis in the project that links an operational metric (delivery outcome) directly to a customer-experience metric (rating), which the purely financial analyses elsewhere in this report do not capture.

7.2.10.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData\pig>notepad pig10_deliverystatus.pig

C:\ECommerceBigData\pig>pig -x mapreduce pig10_deliverystatus.pig
2026-09-11 15:16:31,866 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 15:16:31,869 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 15:16:31,871 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 15:16:32,728 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 15:16:32,728 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789118192716.log
2026-09-11 15:16:33,724 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 15:16:33,827 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 15:16:33,827 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 15:16:34,606 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig10_deliverystatus.pig-2f3176d-3e33-48d2-8d66-176d2d5f14b
2026-09-11 15:16:34,607 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 15:16:35,774 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputformat.separator
2026-09-11 15:16:35,795 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 15:16:35,825 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 15:16:35,869 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED:[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByKeyParallelizer, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 15:16:35,974 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699480192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 15:16:36,065 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 15:16:36,085 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 15:16:36,122 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 15:16:36,123 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 15:16:36,263 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 15:16:36,550 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 15:16:36,567 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 15:16:36,575 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 15:16:36,576 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 15:16:36,579 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 15:16:36,584 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 15:16:36,585 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 15:16:36,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999
2026-09-11 15:16:36,596 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
```

7.2.10.2 Show the actual output

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/DeliveryStatusBreakdown/part-r-00000
Failed,49994,2.9927191262951554
Pending,49774,3.0080765058062444
Delivered,750804,2.9993127367462082
In Transit,149551,3.00175191071942
```

File contents

```
Failed,49994,2.9927191262951554
Pending,49774,3.0080765058062444
Delivered,750804,2.9993127367462082
In Transit,149551,3.00175191071942
```

8. Findings and Interpretation

MapReduce Results :

Job	Result
Job 1-Total Order Count	1000123

Job 2-Category-wise Sales	Clothing 4.5, Electronics 1.3 Health 5.7, Home 8.7, Sports 7.2
Job 3-Avg. Order Value by Country	~\$400–406 across all 10 countries
Job 4-Total Profit by Payment Method	~\$32.1M–\$32.4M across all 5 methods

Pig Results :

Analysis	Result
1. Row Count Check	Raw: 1,000,124 · Cleaned: 1,000,123
2. Total Order Count	1000123
3. Category-wise Sales	Samsung,8136060.9, Dell,8109250.9, Fitbit,8102206.1, Sony,8096025.2, GoPro,8083747, JBL,8074264.5, Philips,8067932.9, Xiaomi,8066665.4 Baseus,8064639, Huawei,8057126.3, Anker,8046326.7, Nike,8038814.4, Apple,8012777.9, Logitech,8012292.1, Nikon,8004462.2, Adidas,8002090.9, Canon,7998829, HP,7997523.5, Lenovo,7994543.6, Asus,7981233.5
4. Average Order Value by Country	Italy,405.7, Spain,403.6, Canada,401.9 France,401.7, Belgium,402.7, Germany, 402.6, Australia,404.2, Netherlands,403.2, United States,403.4, United Kingdom,403.1
5. Payment Method Distribution	PayPal,199881,8.0, Apple Pay,199814,8.0, Debit Card,200805,8.1, Credit Card,199410,8.0, Bank Transfer,200213,8.0
6. Monthly Sales Trend	1,85100,3.4,2,77685,3.1,3,85079,3.4,4,82436, 3.3,5,85100,3.4,6,81699,3.2,7,85029,3.4, 8,84799,3.4, 9,82005,3.3,10,84848,3.4,11,81731,3.212,84612,3.4
7. Top 10 Best-Selling Products	Water Bottle Insulated,67853, Vacuum Robot Cleaner,67703 Tennis Balls Pack,67482, Cotton T-Shirt Pack,67219 Gym Gloves,67215, Storage Box Set,67130 LED Strip Lights 5m,67124, Treadmill Folding,67109 Resistance Bands Set,67042, Desk Lamp LED,66928
8. Top 10 Countries by Revenue	Netherlands,4.0, Italy,4.0, Belgium,4.0, Spain,4.0, United Kingdom,4.03, Australia,4.03, United States,4.02, France,4.023, Canada,4.02, Germany,4.01
9. Brand-wise Total Profit	Clothing 4.5, Electronics 1.3, Health 5.7 Home 8.7, Sports 7.2

10. Delivery Status Breakdown	Clothing 4.5,Electronics 1.3Health 5.7, Home 8.7,Sports 7.2
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Comparing the Two Approaches

Both approaches produced consistent results for the analyses that overlapped — order count and country averages — confirming the reliability of the pipeline rather than any coincidence in one implementation or the other. However, Pig required substantially less code — each analysis was a handful of Pig Latin lines compared to three full Java classes (Mapper, Reducer, Driver) per MapReduce job — while still compiling down to MapReduce jobs under the hood, meaning the productivity gain came without any real loss of processing power.

9. Limitations & Future Scope

Limitations

- Run on a single-node, pseudo-distributed Hadoop setup rather than a true multi-node cluster, so real distributed performance gains are not demonstrated here.
- Batch processing only — no real-time or streaming analysis of new orders as they arrive.
- Cleaning only removes rows with missing `order_id` or `total_price_usd`; other data-quality issues, such as outlier pricing or duplicate orders, were not specifically investigated in this project.
- Results are shown as raw text output rather than charts or an interactive dashboard, which limits how easily a non-technical reader can interpret them.
- Each job/script was run manually from the command line rather than through an automated pipeline, so re-running the full analysis after a data refresh currently takes deliberate, repeated effort.

Future Scope

- Introduce Apache Hive for SQL-style querying of the same dataset, which would lower the barrier for non-programmers to run their own ad-hoc queries.
- Migrate analyses to Apache Spark for faster, in-memory processing, particularly useful if the dataset grows well beyond its current size.
- Deploy on a multi-node cluster, or a cloud Hadoop service such as AWS EMR, to demonstrate genuine horizontal scalability rather than a single-machine simulation of it.
- Add a visualisation layer, such as Power BI or Tableau, to make the results easier to interpret at a glance.
- Automate the pipeline with Apache Oozie or Airflow to remove the need for manual, step-by-step execution each time the dataset is updated.

10. Conclusion

This project demonstrated the practical use of HDFS, MapReduce, and Apache Pig to store and analyse a large e-commerce dataset of over one million order records. The dataset was uploaded to HDFS and verified for integrity before any processing began, establishing a reliable foundation for everything that followed. Four MapReduce jobs, written as custom Java Mapper/Reducer/Driver classes, computed the total order count, category-wise sales,

average order value by country, and total profit by payment method. Ten Apache Pig scripts then extended this analysis to payment distribution, monthly trends, best-selling products, top-revenue countries, brand profitability, and delivery performance, using Pig Latin to express the same style of aggregation far more concisely than the equivalent MapReduce code would have required.

The results show that average order value and profit are both fairly evenly distributed across countries and payment methods respectively, and that the two Pig/MapReduce implementations of the same analysis, average order value by country, produced consistent figures, which validates the overall pipeline rather than any single job in isolation. This project reinforces how distributed Big Data tools can turn a large, real-world transactional dataset into structured business insight, and lays a foundation for future extensions using Hive or Spark, or for scaling the same approach to a genuinely distributed, multi-node cluster.

11. Reference

<https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop>